

Database Systems

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Switch off mobile phones during lectures, or put them into silent mode

Mapping EER Model Constructs to Relations

Step8: Options for Mapping Specialization or Generalization.

- Convert each specialization with m subclasses $\{S_1, S_2, \dots, S_m\}$ and generalized superclass C , where the attributes of C are $\{k, a_1, \dots, a_n\}$ and k is the (primary) key, into relational schemas using one of the four following options:
 - Option 8A: Multiple relations-Superclass and subclasses
 - Option 8B: Multiple relations-Subclass relations only
 - Option 8C: Single relation with one type attribute
 - Option 8D: Single relation with multiple type attributes

Mapping EER Model Constructs to Relations

Option 8A: Multiple relations-Superclass and subclasses

- Create a relation L for C with attributes $\text{Attrs}(L) = \{k, a_1, \dots, a_n\}$ and $\text{PK}(L) = k$. Create a relation L_i for each subclass S_i , $1 < i < m$, with the attributes $\text{Attrs}(L_i) = \{k\} \cup \{\text{attributes of } S_i\}$ and $\text{PK}(L_i) = k$. This option works for any specialization (total or partial, disjoint or overlapping).

Option 8B: Multiple relations-Subclass relations only

- Create a relation L_i for each subclass S_i , $1 < i < m$, with the attributes $\text{Attr}(L_i) = \{\text{attributes of } S_i\} \cup \{k, a_1, \dots, a_n\}$ and $\text{PK}(L_i) = k$. This option only works for a specialization whose subclasses are total (every entity in the superclass must belong to (at least) one of the subclasses).

FIGURE 4.4

EER diagram notation for an attribute-defined specialization on JobType.

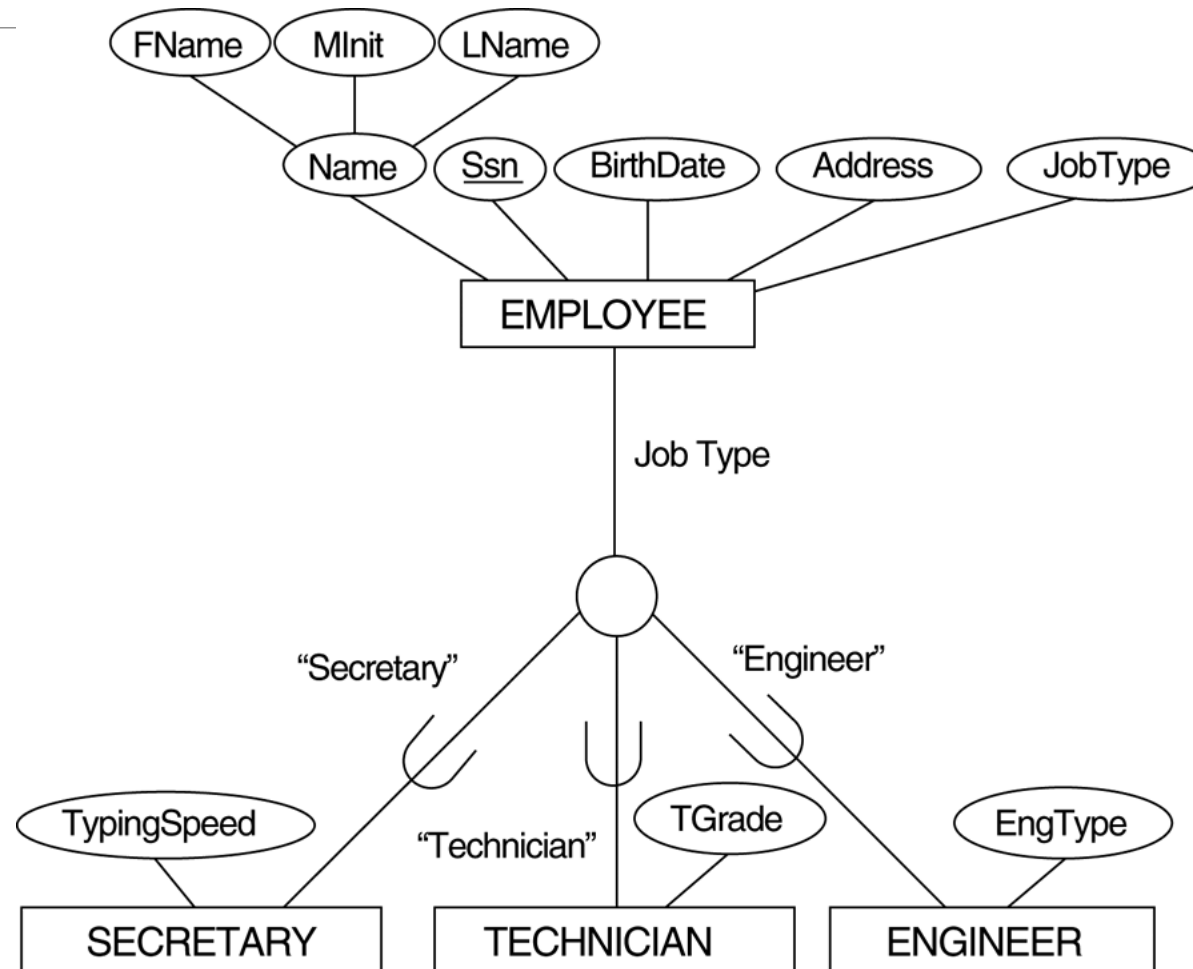


FIGURE 7.4

Options for mapping specialization or generalization.

(a) Mapping the EER schema in Figure 4.4 using option 8A.

(a) EMPLOYEE

<u>SSN</u>	FName	MInit	LName	BirthDate	Address	JobType
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SECRETARY

<u>SSN</u>	TypingSpeed
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TECHNICIAN

<u>SSN</u>	TGrade
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ENGINEER

<u>SSN</u>	EngType
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FIGURE 4.3

Generalization. (b) Generalizing CAR and TRUCK into the superclass VEHICLE.

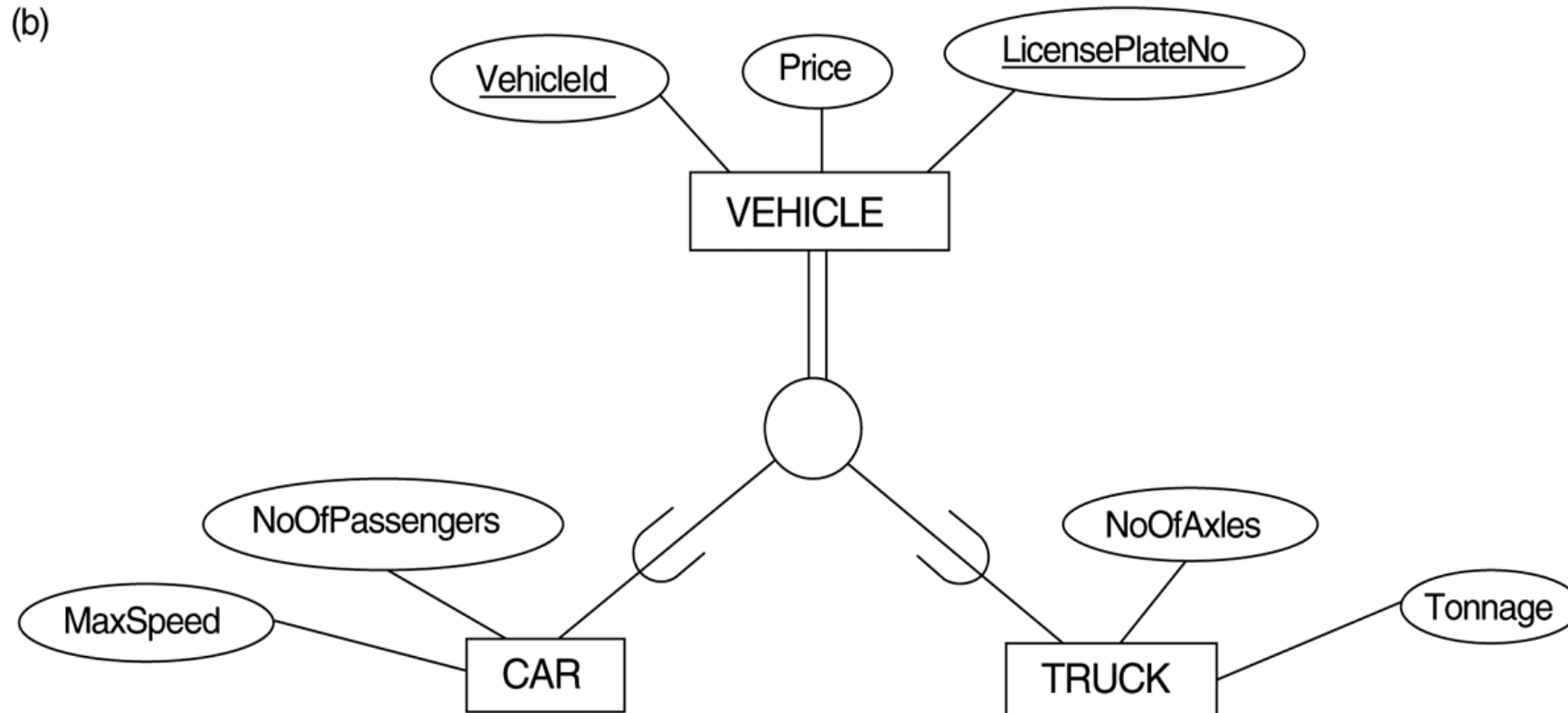


FIGURE 7.4

Options for mapping specialization or generalization.

(b) Mapping the EER schema in Figure 4.3b using option 8B.

(b) CAR

<u>VehicleId</u>	LicensePlateNo	Price	MaxSpeed	NoOfPassengers
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TRUCK

<u>VehicleId</u>	LicensePlateNo	Price	NoOfAxles	
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Mapping EER Model Constructs to Relations (contd.)

Option 8C: Single relation with one type attribute

- Create a single relation L with attributes $\text{Attrs}(L) = \{k, a_1, \dots, a_n\} \cup \{\text{attributes of } S_1\} \cup \dots \cup \{\text{attributes of } S_m\} \cup \{t\}$ and $\text{PK}(L) = k$. The attribute t is called a type (or **discriminating**) attribute that indicates the subclass to which each tuple belongs

Option 8D: Single relation with multiple type attributes

- Create a single relation schema L with attributes $\text{Attrs}(L) = \{k, a_1, \dots, a_n\} \cup \{\text{attributes of } S_1\} \cup \dots \cup \{\text{attributes of } S_m\} \cup \{t_1, t_2, \dots, t_m\}$ and $\text{PK}(L) = k$. Each t_i , $1 < i < m$, is a Boolean type attribute indicating whether a tuple belongs to the subclass S_i .

FIGURE 4.4

EER diagram notation for an attribute-defined specialization on JobType.

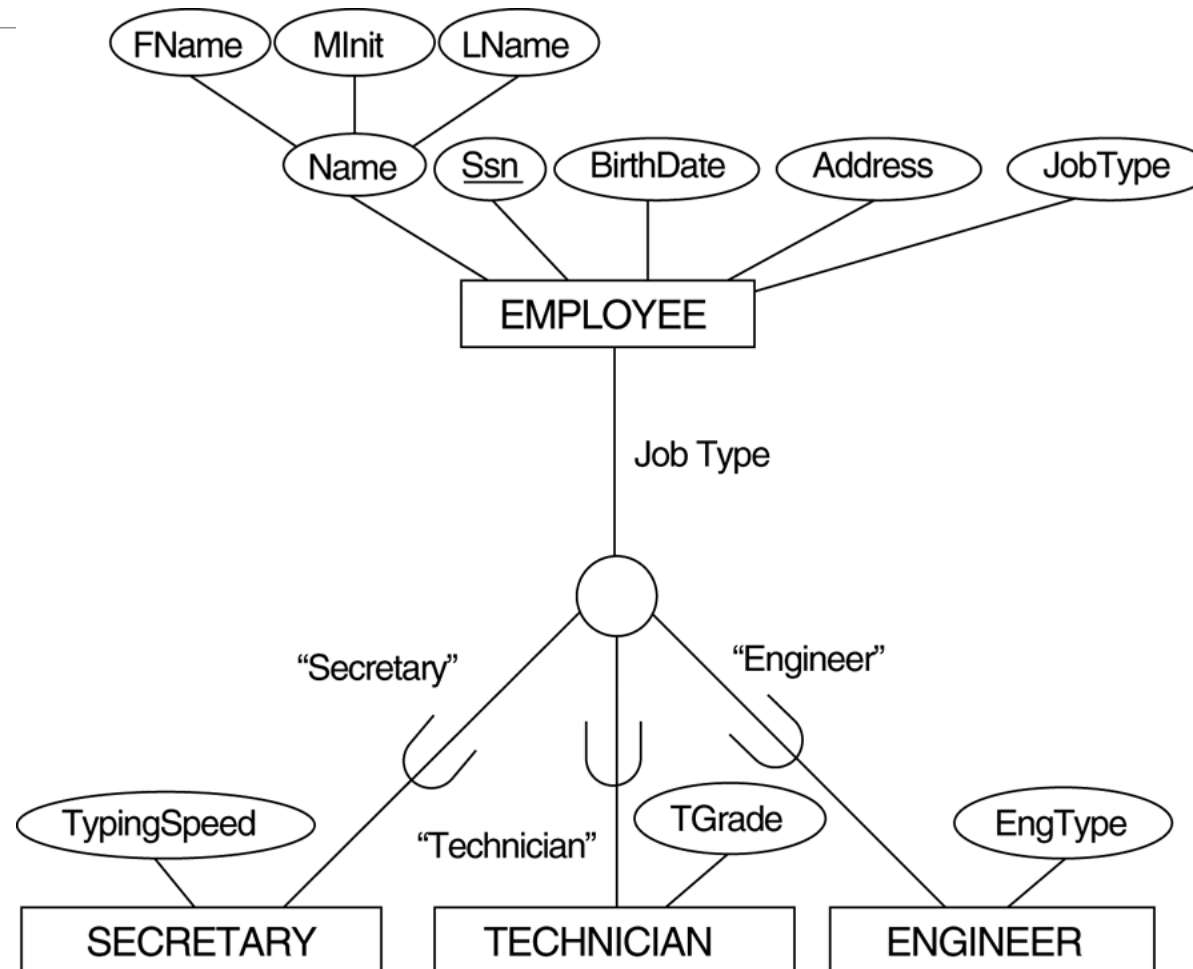


FIGURE 7.4

Options for mapping specialization or generalization.

(c) Mapping the EER schema in Figure 4.4 using option 8C.

(c) EMPLOYEE

<u>SSN</u>	FName	MInit	LName	BirthDate	Address	JobType	TypingSpeed	TGrade	
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FIGURE 4.5

EER diagram notation for an overlapping (non-disjoint) specialization.

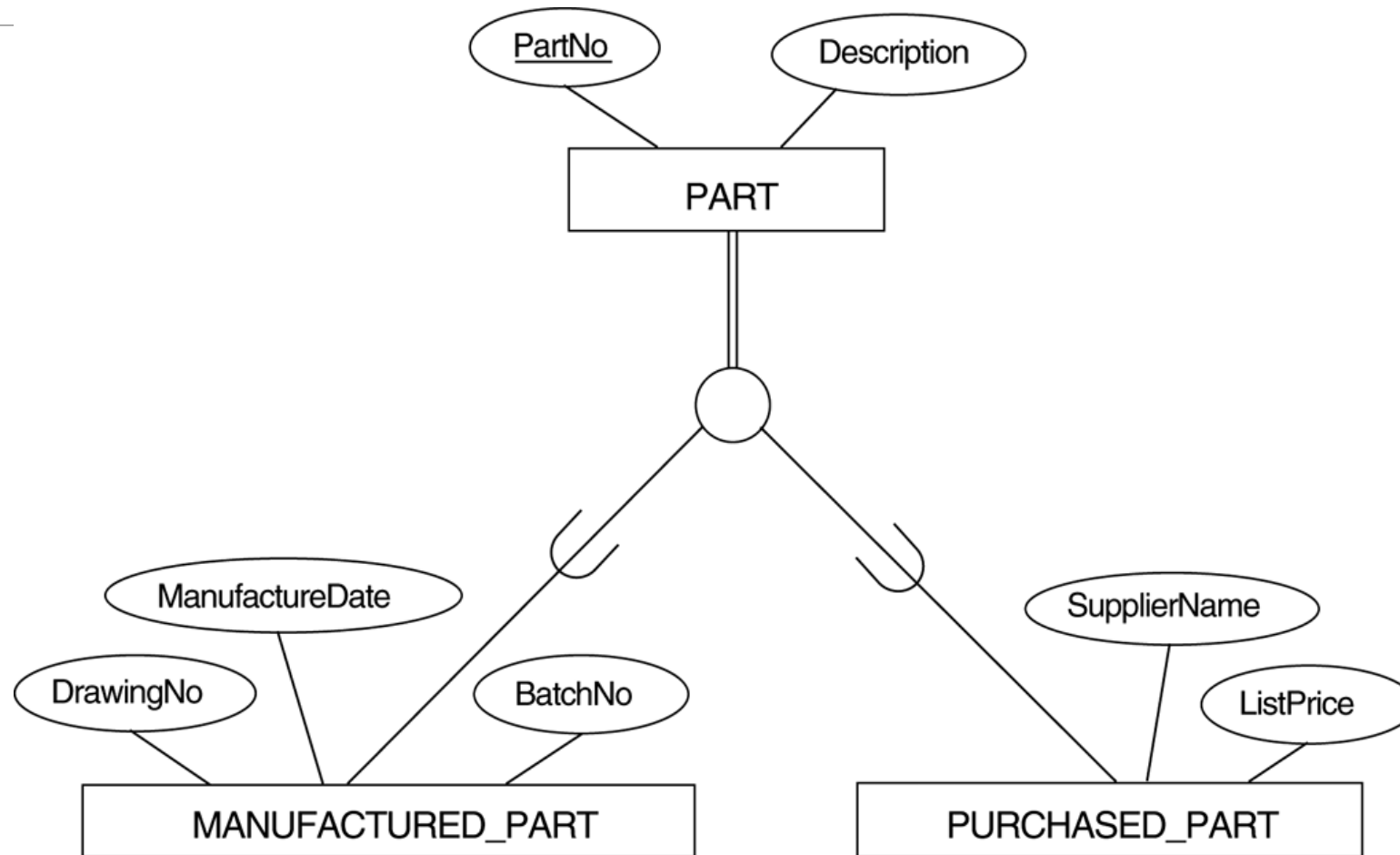


FIGURE 7.4

Options for mapping specialization or generalization. (d) Mapping Figure 4.5 using option 8D with Boolean type fields Mflag and Pflag.

(d) PART

<u>PartNo</u>	Description	MFlag	DrawingNo	ManufactureDate	BatchNo	PFlag	SupplierName	ListPrice
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Mapping EER Model Constructs to Relations (contd.)

Mapping of Shared Subclasses (Multiple Inheritance)

- A shared subclass, such as STUDENT_ASSISTANT, is a subclass of several classes, indicating multiple inheritance. These classes must all have the same key attribute; otherwise, the shared subclass would be modeled as a category.
- We can apply any of the options discussed in Step 8 to a shared subclass, subject to the restriction discussed in Step 8 of the mapping algorithm. Below both 8C and 8D are used for the shared class STUDENT_ASSISTANT.

FIGURE 4.7

A specialization lattice with multiple inheritance for a UNIVERSITY database.

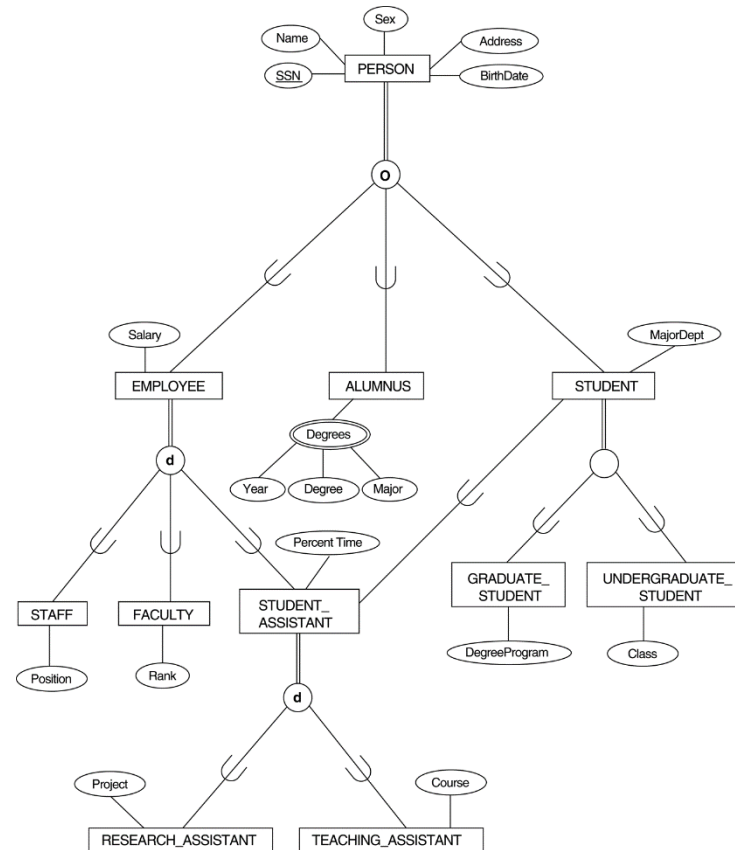


FIGURE 7.5

Mapping the EER specialization lattice in Figure 4.6 using multiple options.

PERSON

<u>SSN</u>	Name	BirthDate	Sex	Address
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EMPLOYEE

<u>SSN</u>	Salary	EmployeeType	Position	Rank	PercentTime	RAFlag	TAFlag	Project	
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ALUMNUS

<u>SSN</u>

ALUMNUS_DEGREES

<u>SSN</u>	Year	Degree	
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STUDENT

<u>SSN</u>	MajorDept	GradFlag	UndergradFlag	DegreeProgram	Class	StudAssistFlag
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Mapping EER Model Constructs to Relations (contd.)

Step 9: Mapping of Union Types (Categories).

- For mapping a category whose defining superclass have different keys, it is customary to specify a new key attribute, called a surrogate key, when creating a relation to correspond to the category.
- In the example below we can create a relation OWNER to correspond to the OWNER category and include any attributes of the category in this relation. The primary key of the OWNER relation is the surrogate key, which we called OwnerId.

FIGURE 4.8

Two categories (union types): OWNER and REGISTERED_VEHICLE.

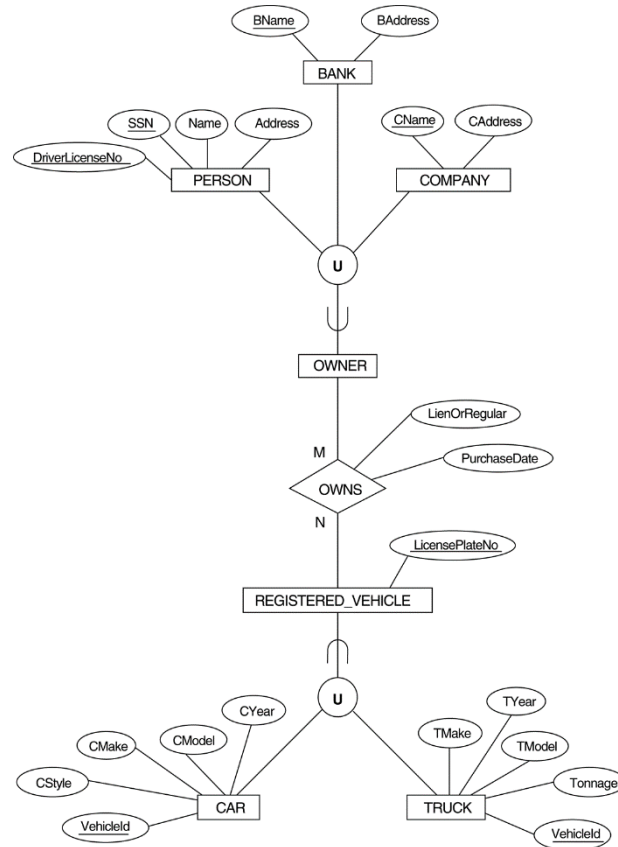


FIGURE 7.6

Mapping the EER categories (union types) in Figure 4.7 to relations.

PERSON

<u>SSN</u>	DriverLicenseNo	Name	Address	
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BANK

<u>BName</u>	BAddress	OwnerId
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COMPANY

<u>CName</u>	CAddress	OwnerId
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OWNER

<u>OwnerId</u>

REGISTERED_VEHICLE

<u>VehicleId</u>	LicensePlateNumber
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CAR

<u>VehicleId</u>	CStyle	CMake	CModel	
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TRUCK

<u>VehicleId</u>	TMake	TModel	Tonnage	TYear
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OWNS

<u>OwnerId</u>	<u>VehicleId</u>	PurchaseDate	LienOrRegular
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